

AMENDMENT NO. 2 NOVEMBER 1996
TO
IS 12709 : 1994 GLASS FIBRE REINFORCED
PLASTICS (GRP) PIPES, JOINTS AND FITTINGS
FOR USE FOR POTABLE WATER SUPPLY — SPECIFICATION
(First Revision)

(Page 1, clause 1.2) — Substitute the following for the existing:

1.2 Provisions relating to fittings fabricated from GRP pipes or by moulding process and joints covered in this standard are for guidance only.'

(Page 2, Table 1) — For 'Max Inside Diameter Range, ID' put '}' against the entries 714, 816 and 918.

(Page 3, clause 7.2, line 1) — Substitute 'effective' for 'nominal'.

(Page 3, clause 7.2.1, line 1) — Substitute 'effective' for 'nominal'.

(Page 3, clause 9) — Substitute the following for the existing:

9 WORKMANSHIP

Workmanship shall be in accordance with good practices as listed in Table 10 and shall meet the acceptance criteria specified.

Table 10 Allowable Defects

Name	Definition	Visual Acceptance Levels
Chip	A small piece broken off an edge or surface	Maximum dimension of break 6.5 mm
Crack	An actual separation of the laminate, visible on opposite surfaces, and extending through the thickness	None
Crack, surface	Crack existing only on the surface of the laminate	Maximum length, 6.5 mm
Crazing	Fine cracks at or under the surface of a laminate	Maximum dimension of crazing, 2.5 mm
Delamination, edge	Separation of the layers of material at the edge of a laminate	Maximum dimension, 6.5 mm
Delamination internal	Separation of the layers of material in a laminate	None
Dry-spot	Area of incomplete surface film where the reinforcement has not been wetted with resin	Maximum diameter, 14 mm
Foreign inclusion (metallic)	Metallic particles included in a laminate which are foreign to its composition	Maximum dimension, 1.5 mm
Foreign inclusion (non-metallic)	Non-metallic particles of substance included in a laminate which seem foreign to its composition	Maximum dimension, 1.5 mm
Fracture	Rupture of laminate surface without complete penetration	Maximum dimension, 29 mm
Air bubble (void)	Air entrapment within and between the plies of reinforcement, usually spherical in shape	Maximum diameter, 3.0 mm
Blister	Rounded elevation of the surface of a laminate, with boundaries that may be more or less sharply defined, somewhat resembling in shape a blister on the human skin	Maximum diameter, 6.5 mm; height from surface not to be outside drawing tolerance

Price Group 1